AXIAL PISTON PUMPS / PRODUCT INFORMATION

P15 Axial Piston Pump

Match the mid-range pump to the simultaneous loads and expected running hours. Use the published curves at matching speed and pressure; separate maximum flow and pressure values do not establish one simultaneous operating point.

The P15 represents a significant step up from the compact P6 range. At 6 kg (13.2 lb) it is the lightest pump in the range capable of industrial-scale flow, and the three-displacement offering gives system designers genuine flexibility to match flow, motor size, and drive power to specific application requirements without changing the pump body or piping.



WaterQuest P15 axial piston water hydraulic pump, 316 stainless steel construction

At a glance

Parameter / value	Parameter / value
Displacements 12/15/19 cc/rev (1.159 in³/rev)	Max Pressure 160 bar (2,321 psi) continuous
Max Flow 37.2 L/min (9.83 US gpm)	Max Input Power 11 kW (14.75 hp)
Weight 6 kg (13.23 lb)	Max RPM (Boosted) 2000 rpm

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Parameter	12 cc/rev (0.732 in³/rev)	15 cc/rev (0.915 in³/rev)	19 cc/rev (1.16 in³/rev)	Config note
Displacement	12 cc/rev (0.732 in³/rev)	15 cc/rev (0.915 in³/rev)	19 cc/rev (1.16 in³/rev)	All configs
Max Continuous Pressure	160 bar (2,321 psi)			All configs
Max Flow (unboosted, 1800 rpm)	~20 L/min (5.28 US gpm)	~25 L/min (6.6 US gpm)	~31 L/min (8.19 US gpm)	Unboosted PC/PA
Max Flow (boosted, 2000 rpm)	~23 L/min (6.08 US gpm)	~29 L/min (7.66 US gpm)	37.2 L/min (9.83 US gpm)	Boosted PB
Max RPM (Unboosted)	1800 rpm			PC / PA
Max RPM (Boosted)	2000 rpm			PB
Max Input Power	~6.5 kW (8.72 hp)	~8.5 kW (11.4 hp)	11 kW (14.8 hp)	At 1750 rpm, 160 bar (2,321 psi)
Max Torque at 160 bar (2,321 psi) (1500 rpm)	~33 Nm (24.3 lbf·ft)	~41 Nm (30.2 lbf·ft)	50 Nm (36.9 lbf·ft)	All configs
Weight	6 kg (13.2 lb)			All displacements
Max Fluid Temperature	50°C (consult WQS above 50°C (122 °F))			
Min Fluid Temperature	2°C (35.6 °F) (antifreeze option available)			
Housing Material	316 Stainless Steel			
P Port (Pressure Outlet)	G3/4" BSPP			Both configs
T Port (Unboosted - tank return)	G3/4" BSPP			PC / PA only
T Port (Boosted - case drain)	G1/2" BSPP			PB only; to sump
Shaft Diameter	18.99 / 18.97 mm (0.748 / 0.747 in)			
Shaft Key	6 x 6 mm (0.236 x 0.236 in)			
Flange / Spigot Diameter	82.55 / 82.50 mm (3.25 / 3.25 in)			
Body Outer Diameter	114 mm (4.49 in)			
Body Length	145 mm (5.71 in)			
Shaft Section Length	43 mm (30 to 35 mm (1.18 to 1.38 in) key zone)			
Shaft Offset	21.50 / 21.30 mm (0.846 / 0.839 in)			
Key Engagement	6.25 mm (0.246 in)			
Base Projection	12 mm (0.472 in)			
Port Face	44 mm (1.73 in) flat-to-flat, 4-hole pattern			

Parameter	12 cc/rev (0.732 in³/rev)	15 cc/rev (0.915 in³/rev)	19 cc/rev (1.16 in³/rev)	Config note
Inlet Filtration Required	10 µm (394 µin) nominal, 25 µm (984 µin) absolute, beta-10 = 75			
Shaft Rotation	PC = CW, PA = CCW / PB = either direction			
Certifications	Confirm applicability and supplied documentation: ATEX II 2GD ck T6, ISO 9001 FM 87247, F4N			

Configuration and ordering data

Code	Motor	Frame	Pressure
P15-019-M05	0.55 kW (0.738 hp)	80 Frame	10 to 20 bar (145 to 290 psi)
P15-019-M08	2.2 kW (2.95 hp)	100L (26.4 US gal) Frame	40 bar (580 psi)
P15-019-M09	4.0 kW (5.36 hp)	112M Frame	60 bar (870 psi)
P15-019-M11	5.5 kW (7.38 hp)	132S Frame	80 to 100 bar (1,160 to 1,450 psi)
P15-019-M13	7.5 kW (10.1 hp)	132M Frame	120 to 140 bar (1,740 to 2,031 psi)
P15-019-M14	11 kW (14.8 hp)	160M Frame	160 bar (2,321 psi)

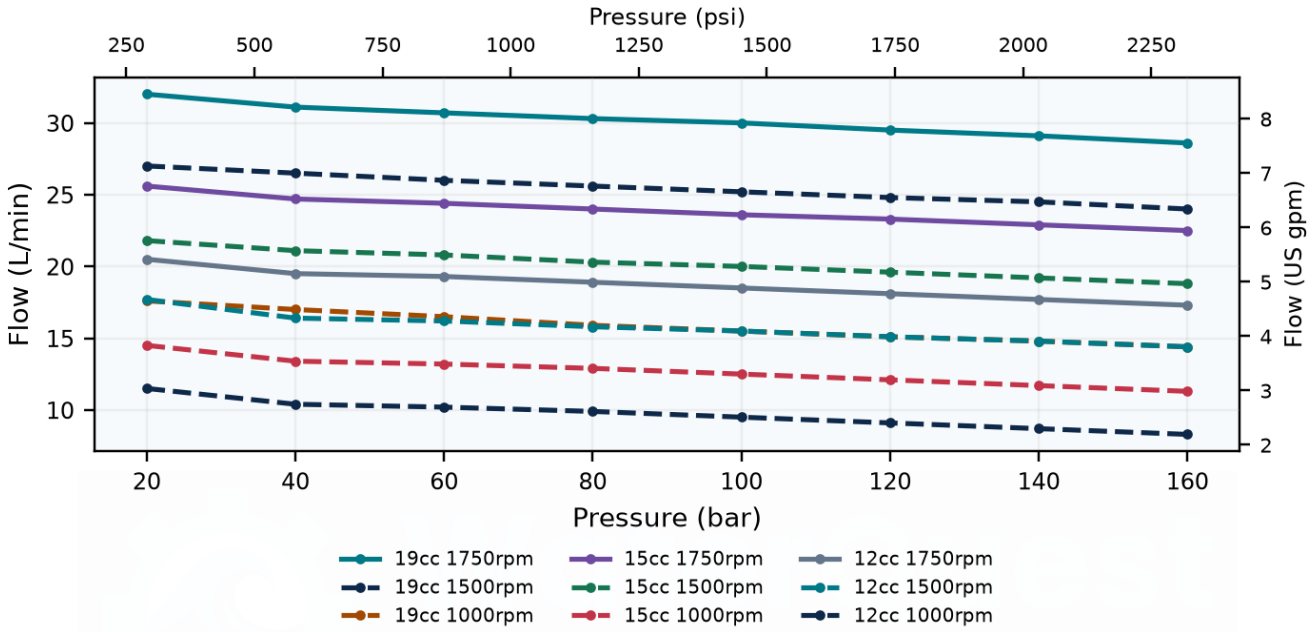
Selection and integration

All performance figures on clean tap water at 15°C (59 °F). The exact operating fluid must be specified on all enquiries and orders. 316 SS Material Test Reports available on request. Consult WaterQuest for motor frame sizing and bell housing selection across all three displacement options.

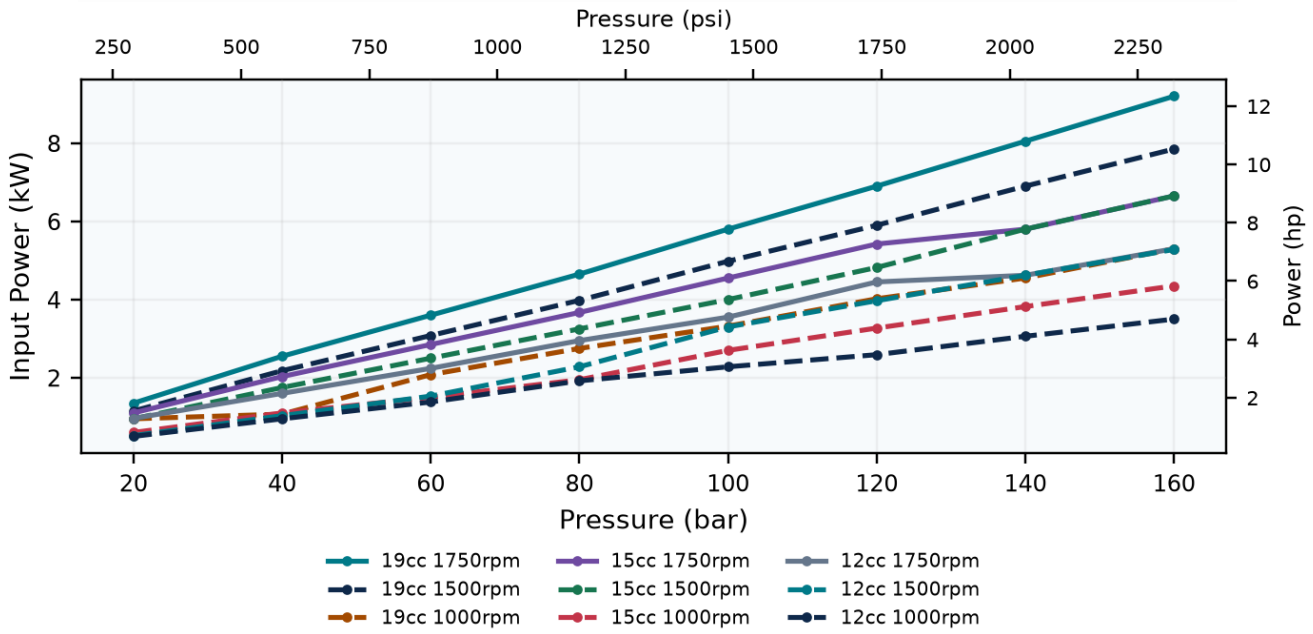
Performance curves

Reference curves reproduced from the product webpage. Curves retain the published data points and speed/displacement labels. Confirm guaranteed performance at the selected duty.

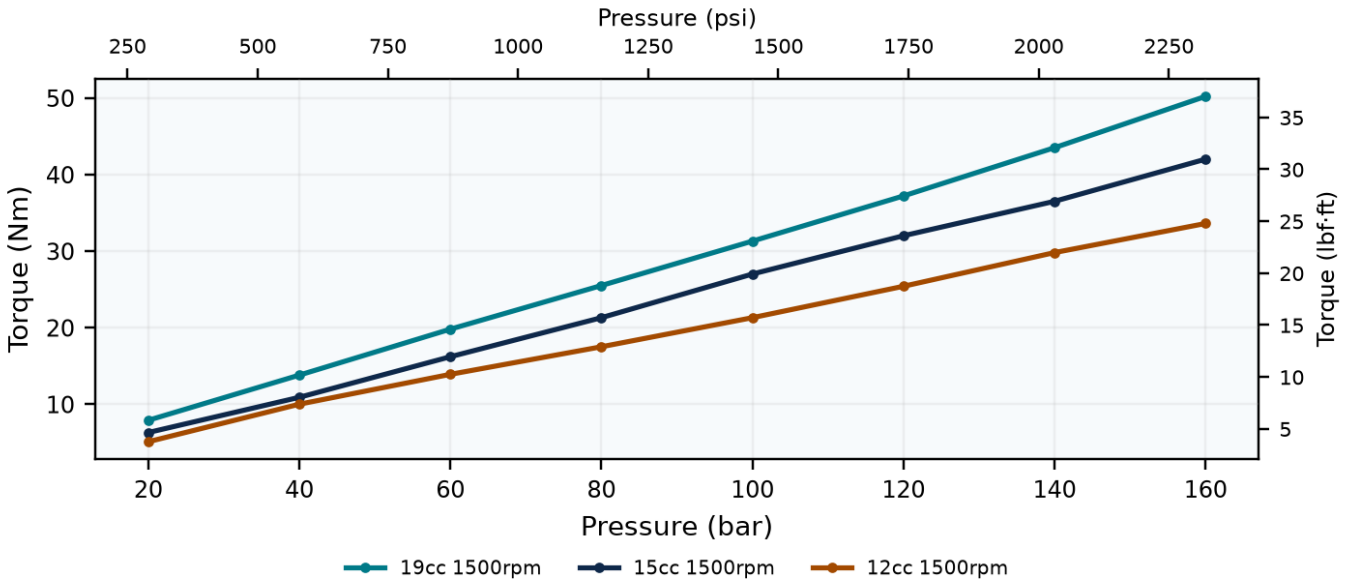
Flow (L/min) vs Pressure (bar)



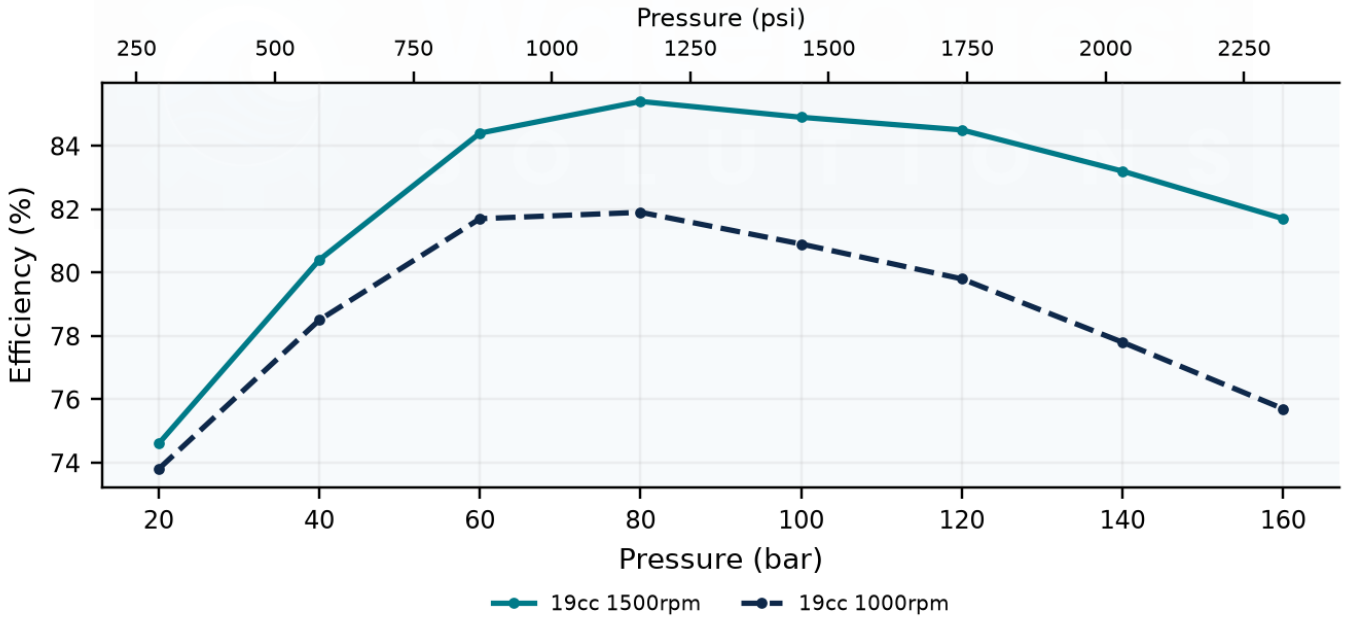
Input Power (kW) vs Pressure (bar)



Torque (Nm) vs Pressure (bar)



Efficiency (%) vs Pressure (bar)

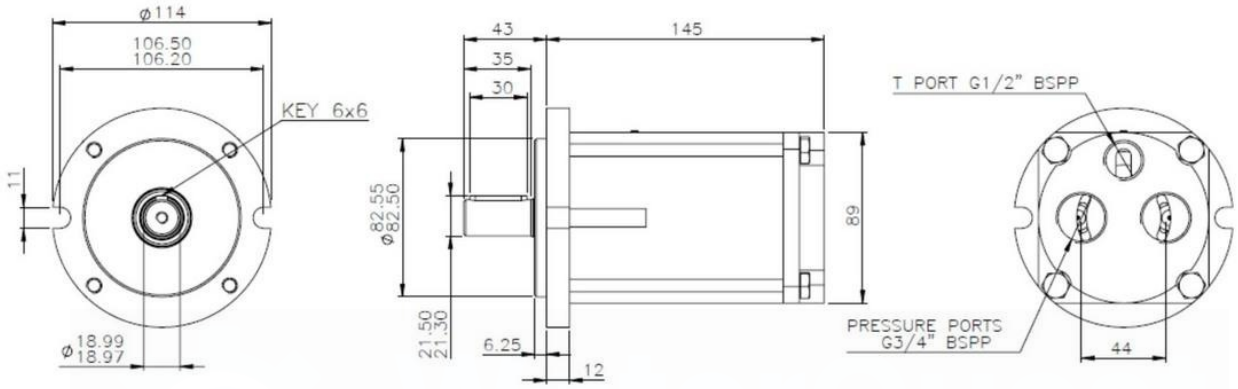


Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

PB160-12W, PB160-15W and PB160-19W boosted P15 dimensional drawing, G1/2 BSPP T port, G3/4 BSPP pressure ports

PB160-12W, PB160-15W & PB160-19W



PC160 and PA160 unboosted P15 dimensional drawing, both ports G3/4 BSPP, all three displacement variants

PC160-12W, PC160-15W & PC160-19W

PA160-12W, PA160-15W & PA160-19W

